Hugo L. Mazzali

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EDUCATION

Cornell University, College of Engineering, Ithaca, NY **Expected May 2027**
Bachelor of Science, Mechanical Engineering | GPA: 3.2
Relevant Coursework: Statics and Mechanics of Solids, Electromagnetism, Thermodynamics, Material Mechanics, Fluids Dynamics, System Dynamics, Mechanical Design, Multivariable Calculus, Differential Equations, Linear Algebra, Mechatronics, Heat Transfer, Automotive Engineering, Simulations and Design

OBJECTIVE

Seeking to apply and build engineering skills at a demanding, driven, and creative internship opportunity this summer

SPECIALIZED SKILLS

Programs: Python, MATLAB, C++, SolidWorks, G-Code, Ansys FEA, GD&T, Microsoft Suite, Adobe Suite,
Mechanical and Test Systems: Wheel Force Transducer, Dynamometer, hydraulic and linear actuators, CNC (3-Axis), Manual Mill and Lathe, Tig Welding, Metal Working and Manufacturing, 3D Printing, QC/FAI processes, Hand tools
Languages: English (fluent), Portuguese (fluent), Spanish (fluent), Italian (intermediate)

ENGINEERING EXPERIENCE

Cornell Baja Racing, Cornell University, NY, Suspension Team Member **Oct 2023-Present**

- Designed and manufactured carbon fiber front and rear tie rods, performing ANSYS ACP and physical testing to validate FOS. Decreased weight by 13% while maintaining legacy FOS of 7.1
- Optimized adhesive bonding process between aluminum and carbon fiber, reducing manufacturing time
- Integrated U-joint and clamped rack interface into steering column reducing steering effort and component wear. Roughly 25% steering effort improvement measured with EMG testing
- Added sinusoidal relation between driver input and steering output to driver spec
- Currently designing and manufacturing front A-arm suspension and manufacturing fixtures
- Machined on-car components, maintained and rebuilt legacy vehicles, serviced vehicle in competition
- Practicing industry level DFM, DFS, and DFA

Corning Incorporated, R&D Mechanical Systems, Corning, NY, Intern **Jun 2022-Sep 2022**

- Designed and built an automated fluid dispensing system for fiber optic manufacturing, integrating pneumatic actuators and closed-loop controls to replace a fully manual process
- Developed custom dependency library to drive actuators, maintaining legacy compatibility
- Wrote G-Code interpreter to interface CNC machinery with automation software
- Independently constructed and operated lab CNC

PROJECTS

Honda CX 500, *Personal Project* **June 2025-Present**

- Full mechanical teardown and redesign of a 1979 motorcycle; machining and welding structural members

Vehicle Dynamics MATLAB Simulator, *Personal Project* **Aug 2025-Present**

- Built dynamic suspension geometry and load transfer model for offroad vehicle

Heat Exchanger Analysis, *Academic Project* **Aug 2025-Dec 2025**

- Built dynamic suspension geometry and load transfer model for offroad vehicle

LQR State Modelling Ball on Beam, *Academic Project* **Aug 2025-Dec 2025**

- Built dynamic suspension geometry and load transfer model for offroad vehicle

CAMPUS INVOLVEMENT

Cornell Film Club, *Co-President*
BRASA Cornell, *Member*
Cornell Footvolley, *Member*
Recreational Soccer Leagues, *Member*

VOLUNTEERING

Habitat for Humanity Restore, Corning, NY
Meals on Wheels, Corning, NY

OTHER INTERESTS: Art and Design / Metalworking / Racing / Filmography / Music / Model making

References Available Upon Request